DEEPAK KATUKURI

AI / ML ENGINEER · COMPUTER VISION · NLP & DEEP LEARNING

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dida Machine Learning GmbH
Hiring Team — Machine Learning Working Student
Berlin Schöneberg, Germany

Application: Machine Learning Working Student

Dear Hiring Team,

Your two example projects read like a map of the work I have already been doing. Inferring roof geometry from 2D satellite imagery is a computer-vision problem close to my intracranial-hemorrhage detector, where I fine-tuned an EfficientNetB0 CNN on RSNA CT scans to 0.98 ROC-AUC and 100% recall. Turning thousands of scanned legal documents into a labelling scheme and an inference task is exactly the shape of my NLP work: a framework-free Retrieval-Augmented Generation pipeline built from scratch in ~270 lines of Python, and a CLIP + DistilBERT multimodal retrieval model trained with InfoNCE contrastive loss to a validation median rank of 10/200. I cover both of dida's pillars — computer vision and NLP — with real, reproducible projects.

What draws me to dida specifically is the research-oriented mindset. I deliberately built my RAG pipeline without LangChain or a vector DB so that every mechanic — chunking, embedding, similarity search, prompt assembly — stays explicit and understandable, rather than hidden behind a framework. That is the same instinct dida describes: using the latest scientific advances to solve real customer problems efficiently, and understanding *why* a method works before shipping it.

On the mathematics: my foundation is in computer science and applied statistics (BTech CSE, and an MSc in Data Science in progress at the University of Europe in Berlin), and I am genuinely eager to deepen the mathematical side of modern deep learning alongside a team with a strong maths and statistics background. My degree and transcripts are attached so you can assess this directly. I code primarily in Python and am comfortable picking up Julia or any tool the problem calls for.

I am based in Potsdam, in the Berlin area, and plan to stay well beyond the next 1.5 years while I complete my Master's. I am available as a working student now for 15–20 hours per week, hybrid in your Schöneberg office, and would move to a full-time role after graduation. I would welcome the chance to contribute to dida's projects and learn from your machine-learning experts.

Thank you for your consideration.

Sincerely,
Deepak Katukuri